

Spin BuAli

Production Readiness Roadmap

Fine-tuning a Persian/English code-switched radiology transcription model, from current pipeline to a deployable, on-device model

Scope: transcription/fine-tuning pipeline only (data → base model → training → evaluation → deployment packaging → pilot). Software hardening of the FastAPI controller itself is out of scope for this document.

Effort ranges are rough planning aids, not commitments — they assume a small focused team and depend heavily on how the incoming city-wide dataset is delivered (raw vs. already paired/verified).

Executive Summary

Spin BuAli's controller already implements three working pipelines (separate STT+reconcile, multimodal audio-LLM, hybrid) against real dictations. Comparing existing outputs across models (Gemini 2.5 Pro/Flash, Gemma-4-12B, Whisper-large-v3 + Aya-Expanse, GPT-5 via a reconcile step) on the same audio files surfaced a clear pattern: only the strongest cloud model (Gemini 2.5 Pro) produces consistently reliable reports on this content. Local/lightweight models fail in identifiable ways — Whisper-large-v3 cannot handle the Persian/English code-switching at all, and Gemma-4-12B both fabricates clinical findings not present in the audio and occasionally loops into repeated text. Across models, exact numeric measurements (stone size, distance to a landmark, cyst dimensions) disagree the most — the single highest clinical-safety risk in this system.

Published precedent shows this is achievable: a Whisper Tiny model has been fine-tuned for Bengali-English code-switched clinical speech with near-perfect accuracy on its test set, and on-device radiology dictation products (e.g. RadMyk) already run entirely on a radiologist's own machine with no GPU. The recommended path here is the same well-precedented recipe: fine-tune a pretrained, Persian-capable Whisper variant on this project's own audio + verified-report data, rather than continuing to rely on prompting off-the-shelf models.

Base model candidates were narrowed to Whisper-family checkpoints that already support Persian (Moonshine, NVIDIA Canary, and Parakeet-TDT were checked and ruled out — none support Persian). The recommended starting point is **vhdm/whisper-large-fa-v1**, a Persian fine-tune of Whisper large-v3-turbo — it already has both Persian phonetics and CPU-friendly speed, so this project's fine-tuning only needs to add code-switching and radiology-domain adaptation on top, not rebuild language support from zero.

Roadmap at a glance

Phase	Focus	Effort
0	Baseline & problem definition	Complete
1	Data acquisition & preparation	4–10 weeks
2	Base model selection & baseline benchmarking	1–2 weeks
3	Fine-tuning	6–10 weeks
4	Correction/reconciliation: prompt first, tune if needed	1–2 weeks (+2–4 if triggered)
5	Evaluation & clinical safety validation	2–3 weeks
6	Deployment packaging	2–4 weeks
7	Pilot rollout	4–8 weeks
8	Monitoring & retraining loop	Ongoing

Best-case total to first production pilot (Phases 1–7, with #3/#4 overlapping): roughly **5–7 months**. Phase 1 (data) is the dominant source of uncertainty — if the incoming city-wide dataset arrives already paired and

radiologist-verified, this compresses; if reports need per-item verification against audio, it stretches well beyond the range above.

Phase 0 — Baseline & Problem Definition

Status: complete

Already done in the course of this project's early evaluation work.

What was established

- Audited the existing controller (separate / multimodal / hybrid pipelines) and its system prompts.
- Compared real outputs from 6+ model/pipeline configurations against the same 4 dictations (Gemini 2.5 Flash/Pro, Gemma-4-12B, Whisper-large-v3+Aya-Expanse, GPT-5 via reconcile).
- Confirmed Gemini 2.5 Pro is the strongest current performer; local Whisper-large-v3 fails on Persian/English code-switching; Gemma-4-12B hallucinates clinical history and can loop into repeated text.
- Identified that numeric measurements (stone size, distances, cyst dimensions) diverge the most across models — the highest-risk failure mode for a radiology report.
- Checked published precedent for on-device, fine-tuned code-switched clinical ASR — confirms the general approach is achievable on realistic hardware.
- Ruled out Moonshine, NVIDIA Canary, and Parakeet-TDT as base models — none currently support Persian.

RISK — Published numbers aren't your numbers

The precedent cited above establishes the approach is achievable in general, not what accuracy this project's own model will hit on its own audio. Treat outside benchmarks as a plausibility signal only, until Phase 5 produces this project's own figures.

Phase 1 — Data Acquisition & Preparation 4–10 weeks

This phase is the actual bottleneck of the whole roadmap. Model choice and training technique matter far less than whether the ground-truth reports are genuinely correct and properly paired to their audio.

Activities

- Acquire the city-wide radiology audio dataset together with its matched final reports.
- De-identify patient information before any audio or text touches external tooling or storage.
- Verify that ground-truth text is the **radiologist-confirmed final report**, not a raw prior STT/LLM draft — training on unverified drafts will teach the model to reproduce today's errors.
- Normalize audio (format, sample rate, channel count) across the whole dataset.
- Stratify the dataset by exam type (abdomen/pelvic, etc.), recording conditions (mic distance, background noise), and speaker, so training/validation/test splits are representative.
- Split by patient/recording session (not by individual report) to avoid the same speaker/session leaking across train and test.
- Carve out a dedicated, measurement-dense test subset — since numeric findings are the top clinical risk, this needs targeted evaluation coverage, not just a random sample.
- Confirm data-usage rights/consent for training on this dataset before work begins.

RISK — Garbage in, garbage out

If the "ground truth" reports are just historical transcriptions rather than independently verified-correct text, fine-tuning will reinforce the same failure patterns already seen in the current pipeline (e.g. hallucinated boilerplate, wrong measurements) — just more confidently.

Phase 2 — Base Model Selection & Baseline Benchmarking

1–2 weeks

Before touching training, establish where each Persian-capable candidate stands on the new dataset, unmodified — this is the number every later phase gets compared against.

Candidates (ranked)

Model	Basis	Params	Role
vhdm/whisper-large-fa-v1	Persian fine-tune of Whisper large-v3-turbo	809M	Primary candidate
openai/whisper-large-v3-turbo	Vanilla OpenAI checkpoint	809M	Control / fallback
MohammadGholizadeh/whisper-large-v3-persian-common-voice-17	Persian fine-tune of full large-v3	1.55B	Accuracy-ceiling reference
aictsharif/whisper-tiny-fa	Persian fine-tune of Whisper Tiny	39M	Lightweight alternative

Activities

- Run all four candidates, unmodified, against a held-out sample of the new dataset.
- Establish baseline Word Error Rate (WER) per candidate.
- Establish a domain-weighted metric — a "Medical Concept WER" that scores errors on clinical terms and numeric measurements more heavily than filler words, since a wrong stone size matters far more than a dropped "uh".
- Compare all four against the current best reference point (Gemini 2.5 Pro, multimodal) to know the size of the gap fine-tuning needs to close.

Phase 3 — Fine-Tuning

6–10 weeks

Adapt the chosen base model(s) for code-switching and radiology vocabulary using the prepared dataset from Phase 1.

Recommended method: staged LoRA to full fine-tune

Stage A - parameter-efficient fine-tuning (LoRA, or QLoRA if VRAM-constrained). Adapters are trained on the attention/projection layers of both the encoder and the decoder, not decoder-only. Decoder-only adaptation is the common shortcut when the problem is purely vocabulary, but part of this project's problem is acoustic (mic distance, background noise, fast/slurred speech), which lives in the encoder, so the encoder needs to adapt too. This stage is cheap and fast, and its job is to validate the whole data-to-training-to-evaluation pipeline end-to-end before committing bigger compute to it.

Stage B - full fine-tuning (all weights unfrozen), once Stage A validates the pipeline and the real dataset volume is confirmed. Full fine-tuning normally risks catastrophic forgetting and overfitting on small datasets, but a city-scale dataset is large enough to justify it, and it reaches a meaningfully higher accuracy ceiling than adapters alone. Use a low learning rate with warmup, and mix a small proportion of general (non-radiology) Persian speech into training batches so the model doesn't lose general Persian ASR ability while specializing.

Activities

- Stage A: LoRA/QLoRA fine-tune on a held-out-validated subset first, to catch data or pipeline problems cheaply before the expensive run.
- Stage B: full fine-tune on the complete prepared dataset once Stage A checks out.
- Train and compare at least the top two candidates from Phase 2 (vhdm/whisper-large-fa-v1 and the vanilla turbo control) at each stage, so the domain fine-tune is judged against a fair baseline.
- Iterate at each stage: train, validate on held-out split, compare checkpoints, repeat.

RISK — Training compute is not deployment compute

Fine-tuning needs GPU compute (cloud or on-prem), even though the final deployed model must run CPU-only on a radiologist's PC. These are two separate infrastructure needs -- sized in the "Infrastructure & Data Needed" section at the end of this document; don't plan training around the target deployment hardware.

Phase 4 — Correction / Reconciliation: Prompt First, Fine-Tune Only If Needed

1–2 weeks (+2–4 more only if triggered)

The controller's existing reconcile step (pipelines.py) already asks an LLM to clean up STT output using a system prompt and a fixed JSON report template. That approach should be tried first, not assumed insufficient — fine-tuning this step is real added cost and should only be spent if prompting alone doesn't clear the bar.

The deciding factor is model size. In the Phase 0 comparison, this exact prompt+template approach worked well on a large cloud model (Gemini 2.5 Pro) but failed badly on a small local model (Gemma-4-12B) on the same task — it fabricated a patient history that didn't exist in the audio and looped into repeated text. Since production deployment needs a small model that runs on the radiologist's PC without a GPU, that failure mode is the one that matters here, and it's not yet known whether prompting alone is enough to avoid it once the STT side is fine-tuned.

Activities

- Take the small, locally-deployable LLM chosen for production and apply the existing reconcile prompt/template, extended with a handful of few-shot examples of the fine-tuned STT model's real output, plus explicit instructions to flag uncertain numeric values instead of guessing.
- Benchmark this prompted-only version in Phase 5 against the fine-tuned STT's actual output.
- Only if it's still hallucinating findings, looping, or getting numeric values wrong at that point: build a (raw STT output → verified-correct report) pair set from real errors and fine-tune the correction LLM on it.

RISK — Don't assume prompting transfers across model size

A prompt/template that works well on a large cloud model is not evidence it will work on the small local model production actually needs — test on the deployment-target model specifically, not on whichever model was convenient during early comparisons.

Phase 5 — Evaluation & Clinical Safety Validation

2–3 weeks

Formal go/no-go checkpoint before any deployment work begins.

Activities

- Benchmark the fine-tuned STT + tuned correction stage against the Phase 2 baselines and the current controller's best pipeline (Gemini 2.5 Pro multimodal) on the held-out test set.
- Run the dedicated numeric-measurement audit from the Phase 1 test subset — this is the single highest-risk area and needs its own explicit pass/fail bar, not just an aggregate WER number.
- Blind radiologist review: a sample of model outputs, reviewed without knowing which model produced them, compared against verified ground truth.
- Stress-test edge cases directly: far/close mic distance, overlapping or very fast speech, rare or unusual terminology.

RISK — Define the bar before you hit it

Agree on a minimum acceptance threshold (e.g. an exact-match rate on numeric findings) before this phase starts. Without a pre-committed bar, it's easy to rationalize "good enough" under deployment pressure.

Phase 6 — Deployment Packaging

2–4 weeks

Turn the validated model into something that actually runs on a radiologist's machine.

Activities

- Quantize the final model (int8 via CTranslate2 / whisper.cpp) for CPU inference.
- Benchmark real-time factor on representative radiologist workstation hardware — not a developer machine — to confirm the earlier CPU speed estimates hold in practice.
- Package the model as a local HTTP service matching the interface the controller already expects at STT_URL (config.py, default localhost:8000) — this is a natural integration point since the controller already treats STT as a pluggable local service.
- Confirm the packaged service works through all three existing pipelines (separate/multimodal/hybrid) without further changes to pipelines.py.

Phase 7 — Pilot Rollout

4–8 weeks

Real-world validation with a small group before any wider rollout.

Activities

- Run a pilot with a handful of radiologists dictating real (new) reports, side-by-side with their current process.
- Capture corrections radiologists make to model output as new, naturally-verified training data.
- Track the same metrics from Phase 5 (WER, Medical Concept WER, numeric-measurement accuracy) in live conditions, not just on the held-out test set.

RISK — Pilot needs real time to be meaningful

Four to eight weeks isn't a fixed number — it's how long it typically takes to gather enough real dictations and radiologist feedback for the signal to be trustworthy. Cutting this short risks shipping on noise.

Phase 8 — Monitoring & Retraining Loop

Ongoing

Production readiness isn't a one-time finish line for a model like this — accuracy needs to be watched and the model needs a path to keep improving.

Activities

- Establish an ongoing process (with consent) for collecting newly corrected reports as future training data.
- Set a periodic retraining cadence using accumulated corrections from real use.
- Version models and keep a rollback path to the previous known-good checkpoint.
- Keep tracking the numeric-measurement accuracy metric specifically in production, not just aggregate WER — it's the metric most tied to clinical risk.

Infrastructure & Data Needed

What needs to be bought, rented, or negotiated before Phase 1 can start. This is the resourcing ask for this roadmap, not a general IT budget — everything below is scoped to what these nine phases specifically require.

Compute (training)

Training needs GPU compute; the deployed model does not. Two distinct training stages need two different GPU tiers, and neither needs to be purchased outright — both fit a short-term cloud rental (AWS, GCP, Azure, RunPod, Lambda Labs, or similar) for the weeks each stage actually runs.

Stage	Purpose	Minimum GPU	Acquire via
Stage A (Phase 3)	LoRA/QLoRA fine-tuning; validates the pipeline cheaply	1x GPU, 16-24GB VRAM (e.g. RTX 4090/3090, or a cloud A10G-class instance)	Cloud rental, days-to-weeks
Stage B (Phase 3)	Full fine-tuning at city-scale data volume	1x GPU, 40-80GB VRAM (e.g. A100 40GB/80GB or H100-class instance)	Cloud rental, weeks; only consider owning if Phase 8 retraining becomes frequent
Deployment (Phase 6+)	Running the finished model on a radiologist's PC	None — CPU only, ~3-4GB RAM footprint for the quantized model	Nothing to buy per seat

Note the asymmetry: the expensive compute is entirely front-loaded into training (Phases 2-4, and periodic Phase 8 retraining). Nothing about deployment requires per-radiologist GPU hardware or ongoing cloud spend.

Data

- The core input: the city-wide radiology audio + matched final-report dataset itself. This requires a formal data-access/data-sharing agreement with whoever holds it — a prerequisite gate before Phase 1 can legally start, and the single most CEO-level item on this list to negotiate.
- Storage: needs sizing once the real dataset scope is known, but plan for tens to low hundreds of GB for audio at this kind of volume, on fast (SSD) storage for training I/O throughput.
- Security: this is patient data — storage needs encryption at rest and access controls regardless of final volume, from the moment it's acquired.
- Consent / regulatory clearance for using patient audio and reports in model training — confirm what's required before Phase 1 begins, not during it.

Minimum audio + labels needed

Based on published Whisper fine-tuning results (domain/low-resource adaptation studies report solid WER gains from as little as 1-8 hours of training audio, improving further with more), scaled up here since code-switched, noisy, fast speech is harder than the mostly single-language cases those studies cover.

Tier	Audio needed	~Recordings*	Gets you
Floor	5-10 hours	~150-300	Pipeline validation (Phase 3, Stage A) — proves the approach works, catches data/setup problems cheaply
Practical minimum	30-50+ hours	~600-1,500	Production-viable quality, ready to face Phase 5 evaluation
Stretch	100+ hours	~2,000+	Where a full fine-tune (Stage B) clearly earns its cost over LoRA alone

*Rough estimate assuming ~1-4 minute recordings, in line with the sample files reviewed for this roadmap — confirm against the real dataset's actual average length once it arrives.

- Diversity matters as much as volume: coverage across radiologists, exam types, and recording conditions (mic near/far, background noise) generalizes far better than the same hour count from one narrow slice.
- Every hour counted above assumes a radiologist-verified label, per Phase 1 — an unverified transcript doesn't count toward this minimum.
- Phase 4's correction-LLM tuning (if triggered) is measured differently and needs far less: a few hundred to ~1-2k verified (STT output → corrected report) text pairs, since it's adapting an already-capable small LLM to a narrow task rather than learning acoustics from scratch.
- Practical implication: the roughly 5-10 hour floor is enough to start Phase 3 Stage A the moment that much of the eventual dataset is in hand — Phase 1 doesn't need to fully complete before training work begins.

People

- ML engineer(s) to run the data pipeline, training, evaluation, and packaging work (Phases 1-6).
- Radiologist time — a genuine resourcing line, not just an engineering one. Needed for verifying ground-truth reports (Phase 1), blind review of model output (Phase 5), and pilot participation (Phase 7). This is very often the actual bottleneck on a project like this, not compute.
- A data-governance/legal contact to negotiate and clear the data-access agreement above.

Tooling (optional, low cost)

- Experiment tracking (e.g. Weights & Biases or MLflow) to compare checkpoints across Phase 3 training iterations.
- A basic model registry/versioning setup for the retraining loop in Phase 8.